

Forgetting-Rate–Bounded Elimination of Nonlinear Recurrences in FEC-Protected Receivers: An HLS Transformation Priced in Decibels

First draft — measured results and research plan. Sections IV–VI report measurements already taken; Section VII is the proposal for the work that follows.

Abstract—High-speed receivers have migrated, generation by generation, from mixed-signal slicing to ADC-plus-DSP architectures: 25 Gb/s per lane became 112, then 224, and OIF is now framing 448 Gb/s per lane, while 5G and 6G fronthaul rides on the same electrical lanes. That migration moved the receiver's difficulty out of the analog front end and into a digital datapath whose design space—tap counts, word lengths, detector order, code choice—is far too large to explore in hand-written RTL. High-level synthesis (HLS) arrived as the vehicle for that exploration, and for feed-forward datapaths it delivers: a C description is scheduled, pipelined, and retimed to an initiation interval of one. *Receivers, however, are not feed-forward.* Their inner loops—decision feedback, add-compare-select, timing and gain recovery, decision-directed adaptation—are recurrences with a quantizer inside the loop, and the recurrence, not the arithmetic, fixes the achievable initiation interval. The classical remedy, look-ahead, is exact for linear recurrences but stops at the quantizer, because a slicer does not commute with the algebra that look-ahead performs. What remains is speculation: enumerate every value the in-loop nonlinearity can take and select afterwards, at a cost exponential in the depth unrolled. The field's current answers either shorten the recurrence (dynamic and speculative scheduling, recurrence-aware placement) or enumerate it exhaustively; none removes it. We propose to remove it, and we take the first step here. We observe that these recurrences *forget*, and we make that quantitative: the forgetting rate $\rho(W)$ is the probability that perturbing the loop state at time $n-W$ changes the output at time n . Measured on a decision-feedback loop, ρ decays geometrically and—the result that motivates everything downstream—the window required to reach a fixed replacement error is *not a design constant*: it spans $W = 1$ to 11 for $\epsilon = 10^{-3}$ across operating points of a single equalizer. Current practice fixes unroll depth and traceback depth as constants, so it is over-provisioned in area or under-provisioned in accuracy, and never both right. We further argue, and show, that such a transformation cannot be judged by the error metrics HLS approximation work uses. At a pre-FEC bit-error rate held exactly at the KP4 RS(544,514) threshold (2.2619×10^{-4} , derived here from the code parameters rather than quoted), post-FEC bit-error rate ranges from 1.7×10^{-14} to 2.262×10^{-4} —more than ten orders of magnitude, and at the upper end the code delivers *no coding gain at all*—depending only on how the errors cluster. Since a recurrence approximation errs by letting stale loop state through, its errors are clustered by construction; mean-squared error and pre-FEC bit-error rate therefore cannot rank two such designs. We propose instead to price the transformation in the receiver's own currency, the SNR required to reach the FEC threshold, against standard-cell area read from a real library. This draft reports the forgetting-rate measurements, the ranking-failure measurement, and the standard-cell costs already in hand; Section VII specifies the synthesis, HLS, and FPGA experiments that follow, together with the conditions under which the hypothesis is to be abandoned.

Index Terms—High-level synthesis, initiation interval, loop-carried dependence, recurrence, decision-feedback equalization, Viterbi algorithm, forward error correction, Reed–Solomon, burst errors, standard-cell area, approximate computing.

I. INTRODUCTION

Two decades of per-lane rate scaling have changed what a receiver *is*. At 25 Gb/s a wireline receiver was a continuous-time equalizer, a slicer, and a short decision-feedback tail. At 112 Gb/s it is an analog front end followed by a time-interleaved analog-to-digital converter and a digital signal processor; published receivers of that class devote roughly half their analog power to the converter alone and run it 64 ways interleaved. At 224 Gb/s the same architecture compensates 31 to 50 dB of insertion loss, and OIF's CEI-448G framework now frames the next generation, where the modulation itself—PAM4 at 212.5 GBd against PAM6 at 170 or PAM8 at 141.6—is still open [23]. Wireless fronthaul rides the same lanes: an O-RAN 7-2x split carries frequency-domain IQ over Ethernet, and the electrical interface underneath it is the same one.

The consequence for design practice is that the hard part of a receiver is now a digital datapath with a very large parameter space. Tap counts, word lengths, the detector's memory order, the placement of floating taps, the interaction with the outer code—these interact, and exploring them in hand-written register-transfer level (RTL) code is prohibitively slow. High-

level synthesis is the answer the field reached for, and for the feed-forward parts of a receiver it is a good one: a C or C++ loop nest over a finite impulse response is scheduled, pipelined, and retimed to an initiation interval (II) of one, and the designer explores the space by editing constants.

A. Where it stops

A receiver is not feed-forward. Table I lists the recurrences that sit at its centre. Each has the same shape: a state updated from its own past, with a *quantizer*—a slicer, a sign, a minimum, a saturation—inside the loop.

TABLE I

Nonlinear recurrences at the centre of a receiver. Each is bound by the recurrence-constrained minimum initiation interval; none is reachable by algebraic look-ahead, because a quantizer sits inside the loop.

Kernel	Recurrence	In-loop nonlinearity
DFE	$d[n] = Q(x[n] - \sum c_k d[n-k])$	slicer Q
Viterbi ACS	$M[n] = \min_j (M_j[n-1] + \lambda)$	minimum
Bang-bang CDR	$\phi[n] = \phi[n-1] + \mu \operatorname{sgn}(e)$	sign
AGC	$g[n] = g[n-1] + \mu \operatorname{sgn}(y - r)$	sign
Layered min-sum	message update over rows	min, sign

Kernel	Recurrence	In-loop nonlinearity
DD-LMS	$\mathbf{w}[n] = \mathbf{w}[n-1] + \mu \mathbf{e}x$	decision device

For a loop whose recurrence path spans d pipeline stages, no new iteration may be initiated until d cycles have elapsed; the recurrence-constrained minimum initiation interval satisfies $\text{RecMII} \geq d$ [9]. Because the loop must close within one symbol period, this is not an abstraction: in the analog instance of the same constraint we measured a StrongARM latch's regeneration time constant at 16.85 ps in a 130 nm process, against a closed-form C_L/g_m of 17.07 ps (1.3 % agreement), and the first decision-feedback tap must resolve, level-shift, and return to the summing node inside a single unit interval. The digital instance is the same statement in a different unit: II, not f_T .

B. Why the classical remedy stops too

For a *linear* recurrence, look-ahead removes the bottleneck exactly. Substituting the recurrence into itself once turns $y[n] = ay[n-1] + x[n]$ into $y[n] = a^2y[n-2] + ax[n-1] + x[n]$, halving RecMII; scattered look-ahead and decomposition extend this to arbitrary pipelining depth at logarithmic overhead [1], [2].

The substitution requires that the loop operator commute with itself under composition. A quantizer does not: $Q(aQ(u)+v)$ cannot be rewritten as a function of u and v through the same algebra. Every entry in Table I is therefore out of look-ahead's reach, and what remains is *speculation*: precompute the datapath for each value the in-loop nonlinearity may take and select the correct branch once the previous decision resolves. This is exactly the loop-unrolled decision-feedback equalizer [7] and the M -step add-compare-select of the parallel Viterbi algorithm [3]. It is exact, and its cost is $|A|^d$ copies of the speculated datapath for alphabet A unrolled d steps—exponential.

C. What we propose, and what this draft establishes

The recurrences of Table I share a property that neither remedy uses: they *forget*. An error injected into the loop state does not persist; its influence on the output decays. Viterbi implementations already exploit one instance of this—survivor paths merge, which is why a traceback depth of roughly five constraint lengths suffices—but as a rule of thumb attached to one algorithm, not as a quantity that is measured and then used to size a transformation.

We make it a measured quantity and propose to build a transformation on it:

HYPOTHESIS

H1 (bound). For a recurrence whose in-loop nonlinearity has a finite output alphabet there exists a computable forgetting rate $\rho(W)$ such that replacing the loop by a feed-forward map over a window of W loop inputs incurs a per-sample disagreement probability bounded by $\rho(W)$.

H2 (crossover). The eliminated form costs 2^{BW} table entries against $|A|^d$ speculated datapaths; which wins is predictable from ρ alone, before synthesis.

H3 (relevance). Current HLS tools report $\text{RecMII} > 1$ for every kernel in Table I and cannot perform this transformation; applied at source level it yields II = 1.

H4 (currency). The replacement error must be priced as required SNR at the FEC threshold, not as mean-squared error, because recurrence approximations produce clustered errors and the outer code is orders of magnitude more sensitive to clustering than to rate.

This first draft establishes the measurements that H1, H2 and H4 rest on—Sections IV, V and VI—and specifies in Section VII the synthesis, HLS and FPGA experiments that test H3 and complete the others, including the conditions under which we abandon the hypothesis.

II. RELATED WORK

A. Recurrence-bound initiation interval in HLS

The HLS literature attacks RecMII by scheduling and by architecture. Loop splitting separates iterations whose dependences differ so that the common case pipelines [11]; polyhedral analysis admits dynamic pipelining where the dependence distance is data-dependent [12]; dynamic scheduling, and its hybrids with static scheduling, resolve dependences at run time with load-store queues and rollback [10]; speculative loop pipelining models the paths explicitly [14]; dynamic loop fusion targets loops whose dependences cross [13]. On the architecture side, recurrence-aware placement collapses a multi-stage recurrence path into few registered cycles by co-locating its endpoints [9]. Every one of these *shortens or reorders* the recurrence. None removes it, and none is specific to the case where the loop operator is a quantizer with a small output alphabet.

B. Look-ahead, unfolding, retiming

Parhi's transformations are the exact remedy for linear recurrences and are the standard reference for pipelining recursive filters [1], [2]. Their scope is the limitation we build on, not a shortcoming: the algebra is exact precisely because the loop is linear.

C. Domain-specific speculation

The loop-unrolled (speculative) decision-feedback equalizer [7] and the parallel Viterbi algorithm that breaks the add-compare-select bottleneck by M -step look-ahead [3] are the two canonical instances of exhaustive enumeration in receivers. Reduced-state and delayed-decision variants [27], [28] cut the state count by merging states, which is a *structural* approximation; our forgetting rate is the *temporal* counterpart, and we expect the two to compose.

D. Tables in place of arithmetic

Distributed arithmetic replaces multiplication by table lookup [4], but the stored function is linear in the address and the technique does not touch recurrences. Data-dependent noise-predictive detection in magnetic recording conditions a detector's branch metric on the data pattern [5], [6]; it is the nearest prior instance of "condition on a window instead of computing," and it is realized as a bank of filters, one per pattern—affordable at disk rates, not at receiver parallelism.

E. Error metrics for approximation

Approximate HLS judges a transformation by average error, maximum error, mean absolute percentage error, peak signal-to-noise ratio, or bit-error rate [18]. All five are first-order statistics of the error *magnitude* or *rate*. None is a statistic of the error's *temporal correlation*, which Section V shows is the dominant term once an outer code is present. That the application metric should be preferred to a numerical one is standard advice; that *the* application metric normally used—pre-FEC bit-error rate—is itself insufficient is the point we make.

III. THE FORGETTING RATE

Let a recurrence carry state $\mathbf{s}[n]$ updated as $\mathbf{s}[n] = F(\mathbf{s}[n-1], u[n])$ with output $o[n] = G(\mathbf{s}[n-1], u[n])$, where u is the loop input and G ends in a quantizer with finite output alphabet A .

Definition 1 (forgetting rate). Run two copies of the recurrence driven by the same input, whose states differ at time n_0 by one admissible perturbation of the most recent decision. Then

$$\rho(W) = \Pr\{ o[n_0+W-1] \neq o'[n_0+W-1] \}, \quad (1)$$

the probability over data and noise that the perturbation is still visible at the output W steps later. Both copies feed back *their own* decisions after the perturbation, so ρ measures the loop's own error propagation, not an oracle's.

Why it bounds replacement error. If the state at time $n-W$ is irrelevant to the output at n with probability $1-\epsilon$, then the output at n is, with that probability, a function of the last W loop inputs alone. A feed-forward map over that window therefore reproduces the recurrence except on an event of probability at most $\epsilon = \rho(W)$. We state this as the working bound, not as a theorem: a proof requires care about whether the perturbation used in (1) is the worst case over admissible states, and we treat that as open (Section VIII).

The design rule that follows. Fix a tolerable replacement error ϵ and read off

$$W(\epsilon) = \min\{ W : \rho(W) \leq \epsilon \}. \quad (2)$$

The eliminated form then needs a table of 2^{BW} entries for a B -bit quantized loop input, against $|A|^d$ speculated datapaths for the unrolled form. Both are known before synthesis.

IV. MEASURED FORGETTING RATES

We measured (1) on a decision-feedback loop inside a full link model—channel, continuous-time equalizer, 11-tap feed-forward equalizer, 8-tap DFE—at three operating points, with

4000 independent perturbation trials each. Table II reports $\rho(W)$ and the window (2) at three tolerances.

TABLE II

Measured forgetting rate of a decision-feedback loop. The window needed to reach a fixed replacement error is not a constant: it spans 1 to 11 across operating points of the same equalizer.

Operating point	$\rho(1)$	$\rho(4)$	$\rho(8)$	$W(10^{-1})$	$W(10^{-2})$	$W(10^{-3})$
NRZ, 20 dB, SNR 26 dB	0.0013	0.0005	0.0005	1	1	4
NRZ, 30 dB, SNR 30 dB	0.1500	0.0320	0.0073	2	8	11
PAM4, 20 dB, SNR 38 dB	0.0008	0.0003	0.0008	1	1	1

Two things follow. First, ρ decays, and on the one operating point where the loop is doing substantial work—30 dB of loss, where the feed-forward equalizer cannot cope alone—it decays geometrically from 0.15 to 7.3×10^{-3} over eight symbols. Second, and this is the result the rest of the work rests on, $W(10^{-3})$ takes the values 4, 11 and 1 on three operating points of one equalizer. A design that fixes its unroll depth, or its traceback depth, at a constant is therefore provisioning for one of these and mis-provisioning for the others.

Read against the crossover of H2, this is also where the transformation's limits appear. With a 4-bit quantized loop input, $W = 1$ is trivial, $W = 4$ gives a 16-bit address (65 536 entries, buildable), and $W = 11$ gives a 44-bit address, which is not. *The transformation is conditional, and ρ is what decides.* We regard reporting that boundary as part of the contribution rather than a weakness; a transformation that claimed to apply everywhere would be the suspicious one.

A confound we have not yet separated. A small ρ may mean the loop forgets quickly, or it may mean the loop is contributing little in the first place—an 11-tap feed-forward equalizer ahead of the DFE leaves the DFE little to do at 20 dB. These have the same signature in (1) and opposite engineering meanings. Section VII-C specifies the control that separates them.

V. WHY MEAN-SQUARED ERROR CANNOT RANK THESE DESIGNS

A recurrence approximation errs in a specific way: it lets stale loop state through, and stale state in a decision-feedback loop produces *runs* of wrong decisions rather than isolated ones. The question is whether an outer code cares about the difference. It cares by more than ten orders of magnitude.

We derive the KP4 threshold rather than quote it. For $RS(n, k)$ over $GF(2^m)$ with $t = (n-k)/2$, the post-decoding symbol error rate follows from the tail sum, which folds by $jC(n, j) = nC(n-1, j-1)$:

$$\sum_{j>t} (j/n) C(n, j) p^j (1-p)^{n-j} = p \Pr\{\text{Bin}(n-1, p) \geq t\}. \quad (3)$$

Bisecting (3) for $RS(544, 514)$, $m = 10$, $t = 15$ at a 10^{-15} post-FEC target gives a pre-FEC threshold of 2.2619×10^{-4} , within 6 % of the value used in practice.

Holding the pre-FEC bit-error rate *exactly* at that threshold and varying only the error structure gives Table III.

TABLE III

Post-FEC bit-error rate at a pre-FEC rate fixed at the KP4 threshold 2.2619×10^{-4} . Only the clustering changes. Beyond $t = 15$ symbols a single burst exhausts the code and the coding gain is zero—this row is arithmetic, not modelling.

Error structure	events / codeword	post-FEC BER
random (the assumed case)	—	1.0×10^{-13}
bursts of 2 symbols	0.1230	1.7×10^{-14}
bursts of 4 symbols	0.0615	8.4×10^{-9}
bursts of 8 symbols	0.0308	6.8×10^{-6}
bursts of 16 symbols	0.0154	2.262×10^{-4}
bursts of 40 symbols	0.0062	2.262×10^{-4}

A design-space exploration that ranks by mean-squared error, by maximum error, or by pre-FEC bit-error rate sees these six designs as identical on the last of those and cannot distinguish them at all on the first two. The correct ranking is by required SNR at the threshold, evaluated on the *actual* error positions the implementation produces, and that is what we propose as the objective.

We are explicit that the insufficiency of pre-FEC bit-error rate is known: instrument vendors say so, and the 802.3 record contains contributions on burst sensitivity, interleaving and pre-FEC monitoring [17], [21], [22]. What is not established, and what we claim, is the consequence for *synthesis*: that the objective function of approximate and precision-tuning HLS is, for this class of hardware, measuring the wrong thing by ten orders of magnitude, and that the fix is cheap.

VI. IMPLEMENTATION COST ALREADY IN HAND

Costs are read from the standard-cell library's own views—cell area from the LEF SIZE record, device count from the SPICE view—after mapping with an open synthesis flow, so that the numbers are a reproducible computation rather than an estimate. The flow refuses to report an area when any gate in the mapped netlist is unmatched, because a silently dropped gate biases area downward, which is to say toward whatever the author is arguing.

TABLE IV

Standard-cell cost of receiver blocks in a 130 nm open library. These are the baselines against which the eliminated forms of Section VII will be measured.

Block	area (μm^2)	gates
11-tap FFE + 8-tap DFE	37 666	4 778
FFE + table variant	42 735	—
128-entry inverse ROM	896	112
MLP, 11×8, 96 multipliers	368 026	51 585

One observation from building Table IV is worth recording because it will recur in the synthesis experiments of Section VII: a table whose contents are trivial is optimized away. An identity-valued ROM mapped to $175 \mu\text{m}^2$ and 7 gates because the synthesizer resolved it to wires, while the same ROM holding real coefficients mapped to $896 \mu\text{m}^2$ and 112 gates. Any area reported for an eliminated recurrence must therefore be measured with the *trained* table contents, never with a placeholder.

VII. RESEARCH PLAN

This section is the proposal. It states, for each remaining claim, the experiment, the platform, the controls, and the outcome that would end the line of work.

A. Stage 1 — Separate forgetting from irrelevance

The confound of Section IV must be removed before anything is built. *Experiment*. Re-measure $\rho(W)$ with the DFE's contribution swept independently of the operating point, by (i) disabling the feed-forward equalizer so the DFE carries the whole postcursor, (ii) sweeping the DFE tap energy Σc_k^2 as an independent variable, and (iii) reporting ρ against that energy rather than against loss and SNR. *Control*. A loop with all taps forced to zero must give $\rho(1) = 0$ identically; any non-zero value is measurement leakage and invalidates the instrument. *Kill condition*. If ρ tracks tap energy so tightly that it carries no information beyond Σc_k^2 , then ρ is not a new quantity and H1 collapses to a restatement of tap energy.

B. Stage 2 — Synthesis, both forms, same library

Experiment. Generate RTL for three forms of the same DFE—recurrent, speculative at depth $d = 1, 2, 3$, and eliminated at $W = W(\epsilon)$ —and map all of them through the same open flow to the same 130 nm library. Report cell area, device count, and static-timing critical path. *Controls*. (i) Tables must be populated with trained contents, per Section VI. (ii) All three forms must be verified bit-exact against the same reference vectors in simulation before any area is quoted; an area for a netlist that has not been shown functionally equivalent is not reported. (iii) The recurrent form's critical path must be reported through the loop, not the longest combinational path, since those differ. *Kill condition*. If the eliminated form's area at $W(10^{-3})$ exceeds the speculative form's at every operating point in Table II, H2 predicts nothing useful.

C. Stage 3 — HLS: does the tool see it?

Experiment. Describe each kernel of Table I in C and synthesize with a commercial HLS tool at a fixed target period. Record the reported II, the reported RecMII, the resource estimate, and the scheduling report's stated reason for $\text{II} > 1$. Then apply the source-level transformation and record the same quantities. *Expected outcome (H3)*. $\text{II} > 1$ for every untransformed kernel with the recurrence named as the cause; $\text{II} = 1$ after transformation. *Controls*. (i) Identical pragmas, target period and tool version across both arms. (ii) A feed-forward FIR of the same tap count as a positive control—it must reach $\text{II} = 1$ in both arms, confirming that nothing else in the harness limits II. (iii) Report tool version explicitly; scheduling behaviour is version-dependent and an unversioned II claim is not reproducible. *Kill condition*. If the tool already achieves $\text{II} = 1$ on any Table I kernel without the transformation, that kernel leaves the study and we say so.

D. Stage 4 — FPGA measurement

Simulation cannot establish that a transformed design closes timing, and it cannot reach the error rates the FEC argument needs. Both require hardware.

Platform requirements. The measurement dictates the board rather than the reverse:

- *Fabric.* A device with hard DSP slices and block RAM, so that the multiplier-free and table-based arms are priced on a fabric where multipliers are *not* free—the opposite of the platform bias we have criticized elsewhere. A mid-range 16 nm-class part (e.g. Zynq UltraScale+ or Agilex-class) is sufficient; the claim is about II and relative resource, not about absolute frequency records.
- *Capacity.* Enough block RAM to hold the largest table under test. At $W = 4$ and $B = 4$ the table is 65 536 entries; at $W = 5$ it is 2^{20} , which sets the practical ceiling and must be reported as such.
- *Instrumentation.* On-chip error counting at line rate, since off-chip capture cannot sustain the bit counts required. A pattern generator, the datapath under test, and a comparator with per-codeword symbol-error histogramming—not merely a bit-error counter, because Section V is precisely the argument that a bit-error counter is insufficient.
- *Observability.* The histogram, the error-position stream for offline post-FEC evaluation, and the II and $f_{\max}$ reported by place-and-route, not by synthesis estimate.

Statistical requirement. To distinguish a post-FEC rate of 10^{-9} from 10^{-6} with confidence, the run must accumulate enough codewords that the rarer event is observed tens of times; at 10^{-6} codeword failure rate and 5440 bits per codeword this is order 10^{11} bits per operating point. At a realistic $100 \text{ MHz} \times 64\text{-bit}$ datapath that is minutes per point, which is why the counting must be on-chip. Operating points below that rate are reported as bounds, not as values.

Controls. (i) A floating-point software reference must reproduce the FPGA's error *positions*, not merely its error rate, on a common seed and a common injected noise sequence—if the positions differ the two are not the same experiment. (ii) Each arm is run at identical SNR, identical seed, and identical pattern. (iii) An unmodified recurrent arm is measured on the same board in the same session, so that board, temperature and toolchain are held fixed. (iv) Seeds are reported individually, not averaged, and a claim is made only where it holds on every seed.

Instrument budget. Oscilloscope for clocking and timing-closure debug only; no analog claim is made from it. No RF, no anechoic environment, and no wireless measurement is required for any claim in this work—the study is entirely a baseband digital one, and we say so rather than allowing a reader to assume otherwise.

E. Stage 5 — The other kernels

Only the decision-feedback loop is measured here. The remaining rows of Table I are the generalization test, and they differ in ways that matter: the add-compare-select recurrence forgets by path merging with a well-understood geometric rate, so it is the case where a *predicted* W can be compared against a measured one; the bang-bang timing loop is deliberately slow to forget, since its whole purpose is to integrate, and we expect it to fail the crossover—which makes it the honest negative example the study needs.

F. Deliverable

A source-level transformation, expressed as an HLS pragma and a supporting library, that takes ϵ as a parameter, measures ρ from a calibration run, selects between recurrent, speculative and eliminated forms by the crossover of H2, and reports the cost of its choice in both μm^2 and decibels at the FEC threshold.

VIII. THREATS TO VALIDITY

- *The bound in Section III is a working bound.* Definition 1 perturbs the most recent decision by one admissible step. Whether that is the worst case over admissible state pairs is open; if it is not, $\rho(W)$ under-estimates the replacement error and every window in Table II is optimistic.
- *The confound of Section IV-D is unresolved.* Two of the three operating points may be reporting an idle loop rather than a forgetful one.
- *Channels are synthetic.* No measured backplane scattering parameters were used. This bounds the claims about particular operating points, though not the arithmetic of Table III.
- *Table III uses a burst model, not measured error positions.* The $B \geq 16$ rows are arithmetic—a burst longer than t symbols within one codeword is uncorrectable by construction—but the event rates that map a given burst length onto a given pre-FEC rate follow from a model, and Stage 4 replaces them with measured positions.
- *Interleaving is not yet modelled.* Standardized links interleave codewords precisely to break bursts. The strength of the Section V argument depends on whether the interleaver breaks the bursts a transformed recurrence actually produces, and that comparison is owed.
- *Bibliographic verification is incomplete.* Several references below were identified through secondary sources because direct access to publishers was unavailable from the environment in which this draft was prepared; they are marked, and must be checked against the published record before submission.

IX. CONCLUSION

The recurrence, not the arithmetic, sets the initiation interval of a receiver's inner loop, and the classical remedy stops at the quantizer that every such loop contains. We propose to eliminate the recurrence rather than shorten or enumerate it, using the fact that these loops forget, and we have made the forgetting quantitative. The measured result that organizes the rest of the work is that the window required for a fixed replacement error is not a design constant: it takes the values 1, 4 and 11 across operating points of a single equalizer, so that any fixed unroll or traceback depth is wrong somewhere. We have also shown why such a transformation cannot be judged by the metrics the approximation literature uses: with the pre-FEC bit-error rate pinned exactly at the KP4 threshold, changing only the clustering moves the post-FEC rate across more than ten orders of magnitude, and at bursts longer than the code's correction radius the coding gain is zero. What remains is to build both forms, to measure what the synthesis

tool does with them, and to close the loop on hardware; Section VII says how, and says what would make us stop.

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